

NISQ VANGUARD

Internship Development Program

Four Major Technical Tasks & Deliverables

This internship is structured around four connected development tasks. Each task produces a real component of the NISQ Vanguard digital ecosystem, culminating in a unified production-ready platform.

Task	Project	Primary Outcome
1	Official Website	Professional responsive NISQ Vanguard website
2	AI Co-Pilot Agent	Intelligent assistant integrated into the website
3	Interactive Labs	Hands-on learning and challenge platform
4	Ecosystem Integration	Unified users, data, AI, labs and administration

TASK 1 — NISQ VANGUARD OFFICIAL WEBSITE

Objective: Build a professional, responsive website representing the complete NISQ Vanguard organization.

Intern Responsibilities / Features:

- Responsive UI for mobile, tablet and desktop.
- Pages: Home, About, Services, Products/Projects, Internships, Courses, Events, Research & Innovation, Labs and Contact.
- Modern navigation, footer and reusable components.
- Authentication and backend/database integration where required.
- Admin/content management functionality.
- SEO metadata and structured content.
- Production deployment.

Expected Deliverable: A fully functional, production-ready NISQ Vanguard website.

Evaluation: UI/UX, responsiveness, code quality, backend integration and deployment.

TASK 2 — NISQ VANGUARD AI CO-PILOT AGENT

Objective: Build an AI-powered NISQ Vanguard Co-Pilot that acts as an intelligent assistant across the website.

Intern Responsibilities / Features:

- Answer questions about NISQ Vanguard.
- Explain products, projects, services, courses and internships.
- Guide users through the website and help locate relevant labs.
- Answer technical and learning questions.
- Generate learning recommendations.
- Search the organization knowledge base.
- Provide contextual answers based on the page being viewed.

- Use an agent architecture with LLM, RAG, knowledge base, website search, database and API/tool layer.

Expected Deliverable: A working AI Co-Pilot integrated directly into the NISQ Vanguard website.

Evaluation: AI accuracy, RAG quality, agent architecture, tool integration, UX and security.

TASK 3 — NISQ VANGUARD INTERACTIVE LABS

Objective: Build an online laboratory platform where students learn through practical challenges.

Intern Responsibilities / Features:

- Cybersecurity labs: Linux, networking, web security, reconnaissance, vulnerability analysis, digital forensics, cryptography and SOC fundamentals.
- Programming labs: C, Python, Java, JavaScript, SQL and DSA.
- AI/Data Science labs: data analysis, machine learning, computer vision, NLP, Generative AI and AI security.
- Student accounts and lab catalogue.
- Difficulty levels, instructions, challenges and automatic validation.
- Scoring, progress tracking, leaderboard, badges and certificates.
- Admin lab creation and management system.
- Minimum demonstration target: 3 complete working labs.

Expected Deliverable: A functional NISQ Vanguard Labs platform with at least three complete demonstration labs.

Evaluation: functionality, challenge quality, automation, UX, scoring and reliability.

TASK 4 — NISQ VANGUARD ECOSYSTEM INTEGRATION

Objective: Integrate all NISQ Vanguard platforms into one unified ecosystem.

Intern Responsibilities / Features:

- Single user account and role-based access for Student, Intern and Admin.
- Central database for users, courses, labs, projects, events, internships, progress, certificates and AI conversations.
- Course/payment integration where applicable, including payment verification and transaction records.
- AI Co-Pilot, knowledge base, RAG and AI-powered search integration.
- Lab progress, scores, challenges, leaderboards and certificates connected to user accounts.
- Contact forms, notifications, email integration and event registration.
- Central admin dashboard for users, courses, labs, events, internships, content and analytics.

Expected Deliverable: A unified NISQ Vanguard platform where all major services share authentication, data and workflows.

Evaluation: integration quality, security, API design, data consistency, testing and deployment.

FINAL INTEGRATED ARCHITECTURE

NISQ Vanguard Main Website → AI Co-Pilot + Interactive Labs + Services/Courses/Events/Internships → Unified User System → Central Database → Admin Dashboard

Recommended Technology Areas

- Frontend: modern responsive web framework, reusable components and accessible UI.
- Backend: secure REST/GraphQL APIs, authentication and role-based authorization.
- Database: structured relational data model with proper validation and audit fields.
- AI: LLM + Retrieval-Augmented Generation (RAG) + tool/API orchestration.
- Labs: isolated, controlled execution/validation environment appropriate to each lab.
- Deployment: Git-based workflow, environment variables, testing and CI/CD.
- Security: input validation, secrets management, access control, logging and secure API practices.

Intern Evaluation Framework

Area	Weight
Official Website Development	25%
AI Co-Pilot Agent	25%
Interactive Labs	25%
Ecosystem Integration & Deployment	25%

Bonus Evaluation

Additional credit may be awarded for clean GitHub repositories, documentation, automated testing, security hardening, responsive design, CI/CD, live deployment and a polished final demonstration.

Internship Outcome

The four tasks are intentionally connected. Interns should not treat them as unrelated mini-projects: each task becomes a building block of the actual NISQ Vanguard platform, and Task 4 brings the components together into a unified ecosystem.